

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Systems and Architectures, Noisy Intermediate-Scale Quantum, Variational Quantum Algorithms, Fault-Tolerant Quantum Computers, Quantum Error Correction

EDUCATION

Sungkyunkwan University (SKKU)

College of Information and Communication Engineering

Bachelor of Science in Electronic and Electrical Engineering

Bachelor of Science in Advanced Semiconductor Engineering

Seoul, South Korea

Expected February 2026

GPA: 3.95/4.0 (4.37/4.5)

System Architecture Track

The Pennsylvania State University

School of Electrical Engineering and Computer Science (EECS)

Exchange Student in Computer Engineering

University Park, PA

Fall 2024

Grade: 4.0/4.0

PUBLICATIONS

SeokHyeon Kong, Dongwhan Kim, Kiwan Meang*, Euseong Seo*, “Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy,” *IEEE Computer Architecture Letters*, 2025. (In Preparation)

SeokHyeon Kong and Ha-young Oh*, “Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system,” *The Journal of the Korean Institute of Information and Communication Engineering*, Oct. 2023

RESEARCH EXPERIENCE

Research Mentee

Remote

IonQ, Advised by Dr. Denny Dahl

08/2025-09/2025

- Solved the Four Corners Map Coloring problem using a Variational Quantum Algorithm.
- Applied graph-theoretic encoding with divide-and-conquer approaches to the Instant Insanity problem on NISQ devices.

Undergraduate Research Intern

Suwon, South Korea

ArchLab @ SKKU, Advised by Prof. Dongmoon Min

07/2025-08/2025

- Academically trained in the field of Quantum / Cryogenic Computer Systems; developed foundational expertise through presentations and defenses.
- Explored cryogenic computer architecture - processors, interconnects, caches, and memory - as well as quantum computer systems with a focus on superconducting qubits.

Undergraduate Researcher

University Park, PA

Penn State CSE, Advised by Prof. Kiwan Maeng

08/2024-09/2025

Curriculum Vitae

- Performed in-depth GPU profiling using Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Conducted Differentially Private Stochastic Gradient Descent (DP-SGD) experiments on multiple models and datasets to characterize system overhead.

Undergraduate Research Intern

Pohang, South Korea

POSTECH QCQN (Quantum Computing and Quantum Networks)

01/2025

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods to enhance ion trapping stability.

HONORS & AWARDS

Quantum Hackathon - Director's Award

2025

President's List (Honor Society at SKKU, **2 consecutive years**)

2024, 2025

Dean's List (**5 consecutive semesters**)

SP2023, FA2023, SP2024, FA2024, SP2025

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000)

FA2024

Student Success Scholarship

2023

TEACHING EXPERIENCE

Academic Tutor

Suwon, South Korea

Sungkyunkwan University (SKKU), Intro to Electromagnetism

09/2023-12/2023

- Taught blue collar workers of Samsung Semiconductor, who study Material-Component Convergence Engineering, at SKKU.
- Primarily focused on problems not covered in class.

International Summer School Teaching Assistant

Seoul, South Korea

Sungkyunkwan University (SKKU), Engineering Mathematics 2

07/2023-08/2023

- Worked as a teaching assistant during the international summer semester.
- Prepared and organized class materials, including exams.

TECHNICAL SKILLS

GPU Profiling (Nsight Systems, Nsight Compute), Circuit (Cadence Virtuoso, SoC, Verilog), Programming (C, Python, CUDA, MATLAB), Quantum Computing (Qiskit), Data Analysis (NumPy, Pandas, Matplotlib), Scientific Writing (LaTeX)

CERTIFICATIONS

Qiskit Global Summer School 2025 Quantum Excellence (IBM Quantum)

CS50's Introduction to Artificial Intelligence with Python (Harvard University)

CS50's Introduction to Cybersecurity (Harvard University)